EcoSort AI

Smart Waste Sorting for a Cleaner Future

Transforming Waste Management

"Ineffective segregation leads to pollution and lost resources.

EcoSort AI fixes this with smart technology, making waste sorting automatic, efficient, and eco-friendly."

The Problem

Manual Sorting is Failing

- **Slow & Labor Intensive:** Current systems cannot keep up with volume.
- **Inaccurate:** 40% of recyclables are contaminated by mixed trash.
- **Hazardous:** Dangerous conditions for sanitation workers.
- **High Carbon Footprint:** Increased landfill usage and methane emissions.

The Solution: EcoSort AI

A smart waste bin and conveyor system powered by advanced AI.

- **Real-time Detection:** Identifies plastic, paper, metal, glass, and organics instantly.
- **95% Accuracy:** Machine learning model minimizes contamination.
- **Auto-Sorting:** Robotic flaps redirect items to the correct bin.
- **IoT Connected:** Smart alerts for fill-levels and collection routes.

How Technology Drives It

AI Vision

High-res cameras & sensors detect material composition in milliseconds.

ML Core

Deep learning model trained on millions of waste images.

Auto-Mechanics

Precision robotic diverters ensure correct sorting every time.

IoT Cloud

Real-time data sync for analytics and fleet optimization.

Target Deployment

Malls & Corporate Offices

Universities & Campuses

Smart Cities & Public Spaces

The Process

1

Drop

User deposits waste item.

2

Scan

Camera captures image.

3

Verify

AI Identifies material.

4

Sort

Flaps divert to bin.

5

Track

Data sent to dashboard.

Data-Driven Insights

Municipalities and businesses gain actionable insights:

- **Live Status:** Monitor bin fill-levels to optimize pickup routes.
- **Impact Reports:** Track total waste processed and CO_2 saved.
- **Compliance:** Verify recycling targets are met with audit logs.
- **Cost Control:** Reduce overflow penalties and operational waste.

Key Benefits

Boosted Recycling Rates Correct segregation increases efficiency by up to 60%.

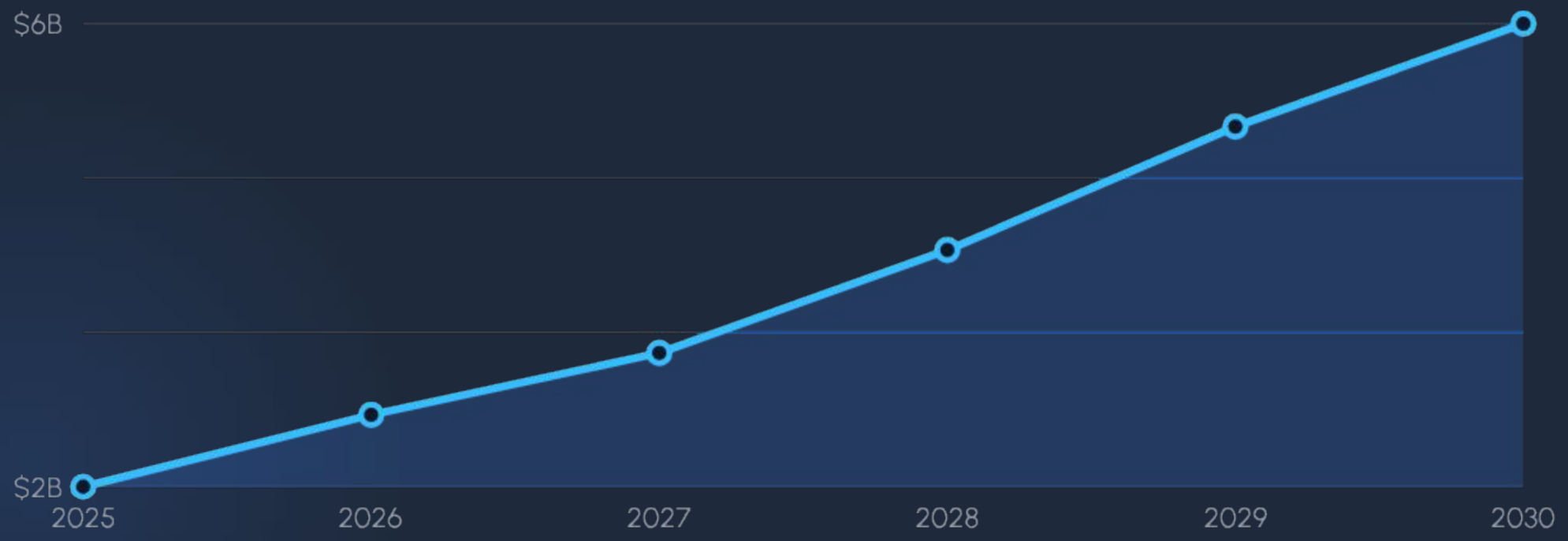
Cost Savings Avoid landfill taxes and generate revenue from clean recyclables.

Safety & Labor Reduces hazardous manual handling for sanitation workers.

Sustainability Goals Directly supports circular economy and corporate ESG targets.

Market Potential

Global Smart Waste Market Projection (Billions USD)



Expected to reach **$4–6 Billion** by 2030 driven by Smart City initiatives.

Revenue Model

Hardware Sales

Direct sales of Smart Bins and Conveyor Systems to large facilities.

SaaS Subscription

Monthly fees for Dashboard analytics, data storage, and AI model updates.

Maintenance

Annual contracts for hardware servicing and sensor calibration.

Enterprise Solutions

Custom installation projects for municipalities and large campuses.

A Cleaner Future

By combining AI, IoT, and automation, EcoSort AI empowers cities to be smarter, safer, and more sustainable.

Let's work together.

Image Sources

https://static.vecteezy.com/system/resources/previews/067/060/147/large_2x/overflowing-landfill-showing-vast-accumulation-of-garbage-and-waste-materials-creating-large-pile-with-varied-composition-free-photo.jpg

Source: www.vecteezy.com

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Source: www.faststreamtech.com

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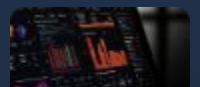
Source: www.sensoterra.com



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